

SIMATS ENGINEERING

ASSESSMENT TOOL 3

Name	Abinеш M
Register Number	192465050
Date	
Time	1 Hr

COURSE NAME: **Cybersecurity Incident Handling And Response**

COURSE CODE: **CSA6001**

COURSE OUTCOME COVERED

C05: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability. (BL3, BL4)*

Activity Tool : Mock Interview (Low)

Assessment Tool Weightage: 10%

Code of Conduct:

I Abinеш M 192465050 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weight age	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
SIEM Data Analysis and Metric Selection	25%	Thoroughly analyzes relevant SIEM and incident data and selects appropriate metrics such as MTTD, MTTR, alert volume, false-positive rate, SLA compliance, and incident recurrence.	Analyzes relevant SIEM data and selects most appropriate metrics with minor omissions.	Performs basic analysis and selects limited metrics with insufficient justification.	Unable to analyze SIEM data or identify appropriate metrics.
Dashboard Design and Visualization	25%	Develops a clear, professional, and interactive dashboard that effectively presents security metrics, incident trends, severity levels, response performance, and key security indicators.	Develops a well-organized dashboard with most relevant metrics and visualizations with minor issues.	Creates a basic dashboard but lacks important metrics, visualizations, or organization.	Unable to develop a meaningful dashboard or present metrics effectively.
KPI Analysis and Incident Response Performance	20%	Accurately analyzes KPIs such as MTTD, MTTR, incident resolution time, false-positive rate, escalation rate, and SLA.	Correctly analyzes most KPIs and identifies major performance trends with minor gaps.	Provides basic KPI analysis but misses important trends or relationships.	Unable to correctly analyze KPIs or evaluate incident response performance.
Continuous Improvement and Recommendations	20%	Identifies security weaknesses and recurring issues and provides specific, prioritized, and actionable recommendations to improve detection, response, SOC processes, and security controls.	Identifies major improvement areas and provides relevant recommendations with minor gaps.	Provides general recommendations with limited connection to the analyzed metrics.	Unable to identify meaningful improvement areas or provide appropriate recommendations.
Documentation, Reporting, and Presentation	10%	Provides clear, professional documentation explaining the dashboard, metrics, findings, trends, recommendations, and continuous improvement actions.	Documentation is well organized and clearly explains most dashboard findings with minor omissions.	Documentation is partially organized with limited analysis or supporting evidence.	Documentation is incomplete, poorly organized, and lacks clear findings or conclusions.

SIEM Metrics, Incident Response Performance, and Dashboard Design

Question 1: SIEM Metrics and Incident Response Performance

Measuring incident response effectiveness requires moving beyond anecdotal impressions of how the SOC is performing and toward a consistent set of quantitative metrics drawn directly from the SIEM platform. As a cybersecurity consultant, my approach is to define each metric precisely, instrument the SIEM and ticketing system to capture the underlying timestamps and outcomes automatically, and then use the resulting data both to evaluate current performance and to drive specific, measurable improvements.

1. Key Metrics and How to Measure Them

Metric	Definition	How to Measure It
Mean Time to Detect (MTTD)	Average time between an incident's actual start and its detection by the SOC	Timestamp of first malicious log event vs. SIEM alert creation time, averaged across incidents
Mean Time to Respond (MTTR)	Average time between detection and the first meaningful response action	Alert creation timestamp vs. ticket acknowledgment/first analyst action timestamp
Incident Resolution Time	Total time from detection through full containment, eradication, and closure	Alert creation timestamp vs. ticket closure timestamp in the tracking system
Alert Volume	Total number of alerts generated over a period, often broken down by severity/source	Direct count from SIEM logs, trended daily/weekly/monthly
False-Positive Rate	Percentage of alerts closed as benign/non-actionable out of total alerts	Count of tickets closed 'false positive' ÷ total tickets, per rule or overall
Incident Recurrence Rate	Percentage of incidents that reoccur after being marked resolved	Count of repeat incidents (same root cause/asset) within a defined window ÷ total resolved incidents

2. Collecting the Underlying Data

- **Timestamp instrumentation:** Ensure the SIEM and ticketing system consistently capture every relevant timestamp — first evidence of malicious activity (from log timestamps), alert generation, analyst acknowledgment, containment, and closure — since every metric above is ultimately a timestamp calculation.

- **Consistent ticket disposition tagging:** Require analysts to tag every closed ticket with a clear disposition (true positive, false positive, benign true positive, duplicate) so false-positive rate and recurrence can be calculated accurately rather than estimated.
- **Root-cause and asset tagging:** Tag each incident with its root cause category and affected asset, which is what allows recurrence to be identified — the same root cause reappearing on the same or related assets within a defined window.
- **Automated data pipeline:** Where possible, feed SIEM and ticketing data into a reporting layer (a BI tool or the SIEM's built-in dashboarding) that recalculates these metrics automatically rather than relying on manual monthly spreadsheet compilation, which is error-prone and quickly falls out of date.

3. Using Metrics to Evaluate and Improve Operations

- **MTTD trending:** A rising MTTD points to detection gaps — missing log sources, under-tuned correlation rules, or insufficient monitoring coverage — and should trigger a review of log ingestion and detection rule coverage.
- **MTTR trending:** A rising MTTR often indicates analyst overload, unclear escalation paths, or alert fatigue from high false-positive volume masking real incidents; it should prompt a staffing, workflow, or alert-tuning review.
- **False-positive rate by rule:** Breaking this metric down per detection rule (rather than only as an org-wide average) identifies specific rules that need tuning, directly reducing analyst fatigue and improving MTTR for genuine incidents.
- **Recurrence rate:** A high recurrence rate for a specific root cause indicates that eradication or root-cause remediation is incomplete — the corrective actions from RCA are not being verified or enforced — and should trigger a review of the closure/verification process itself.
- **Benchmarking over time and against targets:** Tracking each metric monthly against internally set targets (and, where available, industry benchmarks) turns raw numbers into an ongoing performance narrative that justifies further investment in tooling, staffing, or automation where metrics show sustained gaps.

Ultimately, these metrics work best as a connected system rather than in isolation: for example, a team that aggressively tunes down false positives to improve MTTR must also watch MTTD to confirm real threats are not being tuned away along with the noise — reinforcing that SIEM metrics should be reviewed together, not as isolated KPIs, when assessing overall security operations health.

4. Example Metrics Report

A representative monthly snapshot illustrates how these metrics are typically compiled and compared against targets for a management review:

Metric	This Month	Target	Trend / Action
MTTD	38 min	≤ 30 min	Slightly above target; review log coverage

Metric	This Month	Target	Trend / Action
MTTR	22 min	≤ 30 min	On target; no action needed
Avg. resolution time	6.2 hrs	≤ 8 hrs	On target
False-positive rate	34%	≤ 25%	Above target; prioritize rule tuning
Recurrence rate	4%	≤ 5%	On target

Presenting metrics this way — actual versus target, with a clear next action for anything off-target — turns the monthly report into a working management tool rather than a static summary, and creates direct accountability for closing the gaps identified.

5. Common Pitfalls in Metrics Programs

- **Gaming the metric instead of the outcome:** If analysts are measured purely on MTTR, they may prematurely close tickets to stop the clock; pairing MTTR with recurrence rate helps catch this by penalizing incidents that reopen shortly after closure.
- **Averages hiding outliers:** A healthy average MTTD can mask a small number of severely delayed detections on critical assets; tracking percentile metrics (e.g., 90th percentile MTTD) alongside the mean gives a fuller picture.
- **Comparing across incompatible periods:** A spike in alert volume following a new detection rule deployment should not be read as a sudden threat increase; trend charts should be annotated with tooling and rule changes for correct interpretation.
- **Metrics without ownership:** Numbers that no one is accountable for improving tend to stagnate; each metric should have a named owner responsible for investigating sustained deviations from target.

Question 2: SIEM Dashboard Design for Incident Monitoring

When a SOC receives thousands of alerts daily, a well-designed dashboard is what separates a team that can find the handful of alerts that matter from one that drowns in noise. As the SIEM administrator, my design approach prioritizes information hierarchy — showing the most urgent, decision-relevant information first — while keeping deeper investigative detail one click away rather than cluttering the primary view.

1. Dashboard Design Principles

- **Tiered layout:** A top-level summary view for shift leads and management, with drill-down views for individual analysts working specific queues — not a single dashboard trying to serve every audience at once.
- **Severity-first visual hierarchy:** Critical and High severity items are visually prominent (color, size, position) so they cannot be missed even during a fast glance, while Low severity items are summarized rather than itemized.
- **Actionable over decorative:** Every chart should answer a specific operational question (What needs attention right now? Are we meeting our SLAs? Is alert volume trending in a concerning direction?) rather than displaying data for its own sake.
- **Real-time plus trend views:** Combine a live current-state view (what's open right now) with historical trend charts (how are we doing over time), since both are needed for different decisions — immediate triage versus program-level improvement.

2. Key Metrics to Display

- Open incident count by severity (Critical/High/Medium/Low), updated in real time.
- Current SLA status — how many open tickets are within SLA, approaching breach, or already breached.
- MTTD and MTTR, shown both as current rolling averages and trended over recent weeks/months.
- Alert volume trend, ideally broken down by source/rule to spot sudden spikes from a specific detection.
- False-positive rate trend, to monitor detection tuning effectiveness over time.
- Top alerting rules/sources, highlighting which detections are generating the most volume for tuning prioritization.

3. Recommended Dashboard Layout

Dashboard Zone	Content	Chart Type
Top banner	Open Critical/High count, SLA breaches, current on-call analyst	Large number tiles / KPI cards, color-coded
Priority queue	Live list of open incidents sorted by priority and SLA countdown	Sortable table with color-coded SLA timer

Dashboard Zone	Content	Chart Type
Severity breakdown	Distribution of currently open incidents by severity	Donut or stacked bar chart
Volume and trend	Alert volume over time, with anomalous spikes highlighted	Line/area chart, daily and weekly views
Performance metrics	MTTD, MTTR, resolution time, false-positive rate trends	Multi-line trend charts with target thresholds marked
Top sources/rules	Highest-volume alert rules and sources for tuning focus	Horizontal bar chart, ranked

4. Severity Levels and SLA Status Indicators

- **Color-coded severity:** Consistent color mapping across the entire dashboard (e.g., red = Critical, orange = High, yellow = Medium, blue = Low) so analysts build fast pattern recognition rather than reading labels each time.
- **SLA status as a traffic-light system:** Green (within SLA), yellow (approaching SLA deadline, e.g., within 25% of remaining time), and red (SLA breached) — displayed per ticket in the priority queue and aggregated as a count in the top banner.
- **Escalation flags:** A distinct visual marker for tickets that have been escalated or reassigned, so shift leads can quickly see which incidents required additional intervention.

5. Incident Trends to Support Decision-Making

- Week-over-week and month-over-month incident volume, to identify whether overall threat activity or noise is increasing.
- Recurrence trend by root-cause category, surfacing whether specific remediation efforts are actually reducing repeat incidents.
- Time-of-day / day-of-week heatmap of incident volume, useful for staffing and shift-coverage decisions.
- Category-based trend lines (phishing, malware, unauthorized access, etc.), showing which incident types are rising and may warrant additional preventive investment.

6. Supporting Effective Decision-Making

The overall goal of the dashboard design is to let a shift lead answer three questions within seconds of glancing at the screen: What needs my immediate attention? Are we on track against our SLAs? Is anything trending in the wrong direction that needs a longer-term fix? Layering real-time triage data, SLA status, and longer-term trend charts into a single, well-organized view — rather than requiring analysts to run manual queries to answer these questions — is what allows a SOC handling thousands of daily alerts to consistently focus attention on what actually matters.

7. Role-Based Dashboard Views

- **Analyst view:** Focused on their personally assigned queue — open tickets, SLA countdowns, and recent comments/handoff notes — filtered to just their workload rather than the entire SOC's volume.
- **Shift lead / team lead view:** The full priority queue across all analysts, current staffing/coverage, and any tickets at risk of SLA breach needing reassignment, supporting real-time workload balancing.
- **Management / executive view:** High-level trend charts (incident volume, MTTD/MTTR over time, category breakdown) without individual ticket detail, supporting resourcing and investment decisions rather than day-to-day triage.
- **On-call / after-hours view:** A simplified, high-contrast view showing only Critical/High items requiring immediate action, optimized for quick mobile or remote access during off-hours paging.

8. Alerting and Notification Design

- Push notifications (SMS, Slack/Teams, paging systems) triggered automatically for new Critical severity incidents, rather than relying on someone to notice the dashboard.
- Escalating notification cadence for tickets approaching SLA breach — an initial reminder, followed by an escalation to the team lead if no action is taken within a defined window.
- Digest-style (non-urgent) notifications for Medium/Low severity trends, such as a daily summary rather than real-time alerts, to avoid notification fatigue for non-urgent items.
- Clear visual and notification distinction between a new incident and an update to an existing one, so analysts aren't repeatedly interrupted by routine status changes on tickets they're already tracking.

9. Maintaining Dashboard Relevance Over Time

A dashboard designed once and never revisited gradually drifts out of sync with how the SOC actually operates. I recommend a quarterly review cycle where the SIEM administrator, together with SOC leadership, checks whether displayed metrics still reflect current priorities, retires charts nobody references, and adds new views in response to changing threat patterns or newly onboarded log sources — keeping the dashboard a living decision-support tool rather than a static report.

Conclusion

Both scenarios describe two sides of the same discipline: turning raw SIEM data into meaningful operational insight. Defining and tracking metrics like MTTD, MTTR, resolution time, alert volume, false-positive rate, and recurrence rate gives the organization an objective way to measure whether its incident response process is actually improving over time. Designing a dashboard that surfaces those same metrics — alongside real-time severity, SLA status, and trend information — is what makes that data usable in the moment, not just in a monthly report. Together, disciplined metrics tracking and thoughtful dashboard design transform an overwhelming daily alert volume into a manageable, continuously improving security operations program.

Comparing the Two Scenarios

Aspect	Question 1: Metrics Program	Question 2: Dashboard Design
Primary focus	Defining and analyzing performance metrics over time	Presenting real-time and trend data for fast decisions
Primary audience	Security leadership evaluating program performance	Analysts and shift leads triaging in real time
Time horizon	Monthly/quarterly trend analysis and benchmarking	Real-time status plus rolling trend views
Success measure	Metrics improving steadily against defined targets	Analysts consistently acting on the right incidents first

The two are inseparable in practice: a dashboard is only as useful as the metrics behind it are well-defined and accurately measured, and a well-designed metrics program only drives improvement if the data is presented in a way analysts and leadership can actually act on day to day. Investing in both together is what turns SIEM data from a passive log repository into an active driver of security operations performance.